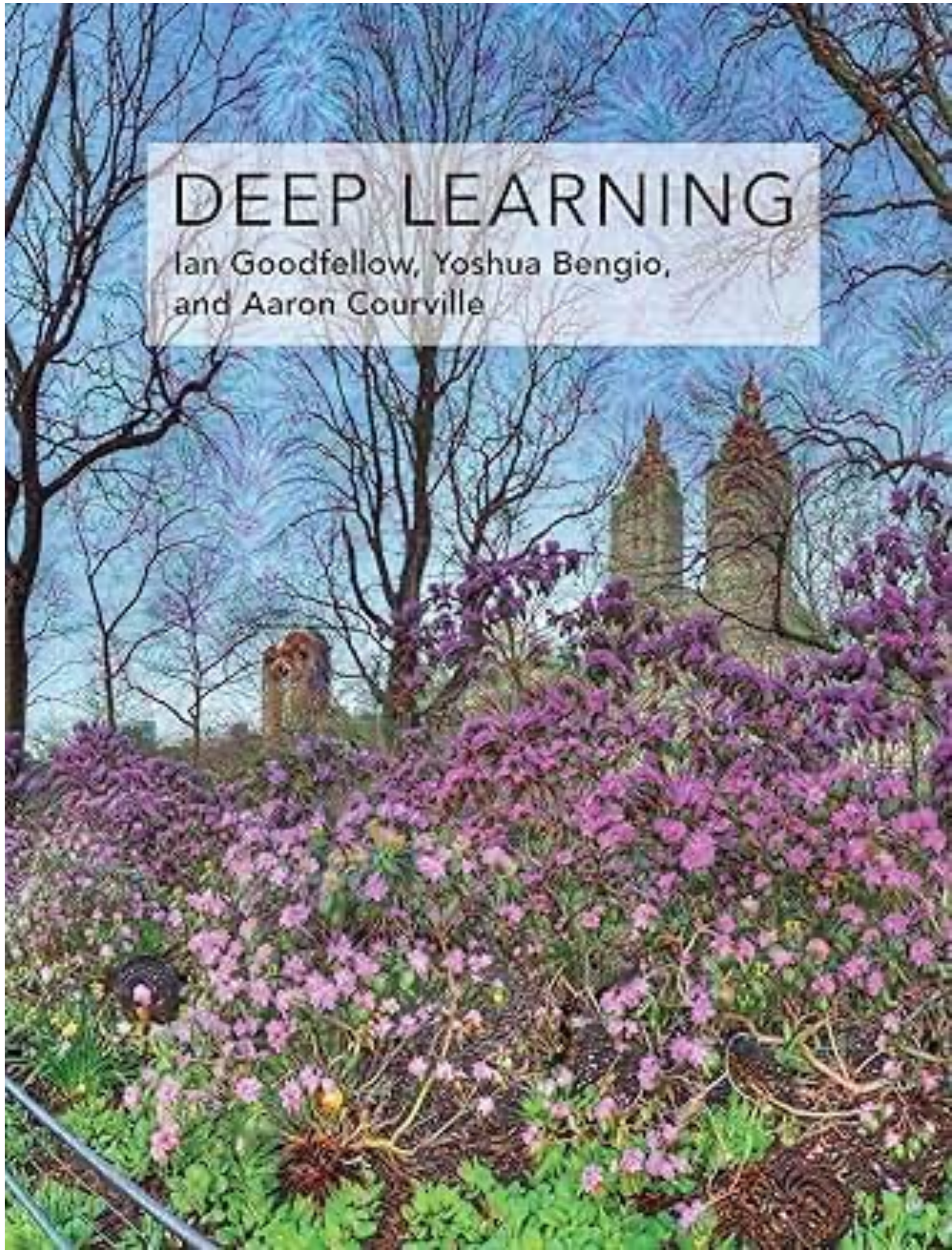
CSCI 6032: Lecture 1 What is Machine Learning

Instructor: Dr. David A. G. Harrison

Things I want to cover today…

- Syllabus
- Tools we will use
- What is machine learning?



syllabus

See handout.

Also available in blackboard.

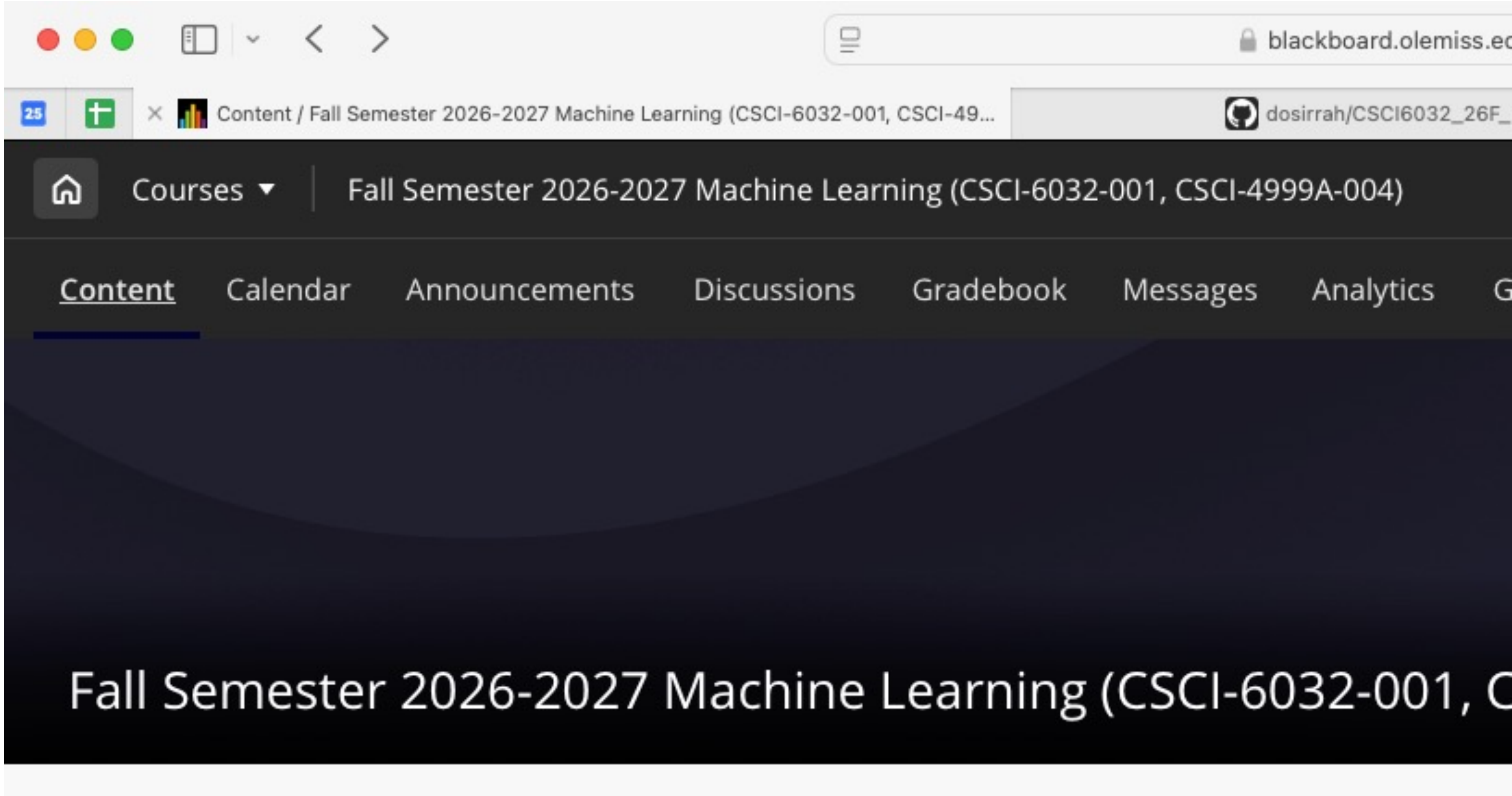
Office Hours

207 Weir Hall

Monday	11:00 AM-12:00 PM
Wednesday	1:00-2:00 PM

Blackboard

Slides and lecture notes will be posted on blackboard and in github.



github

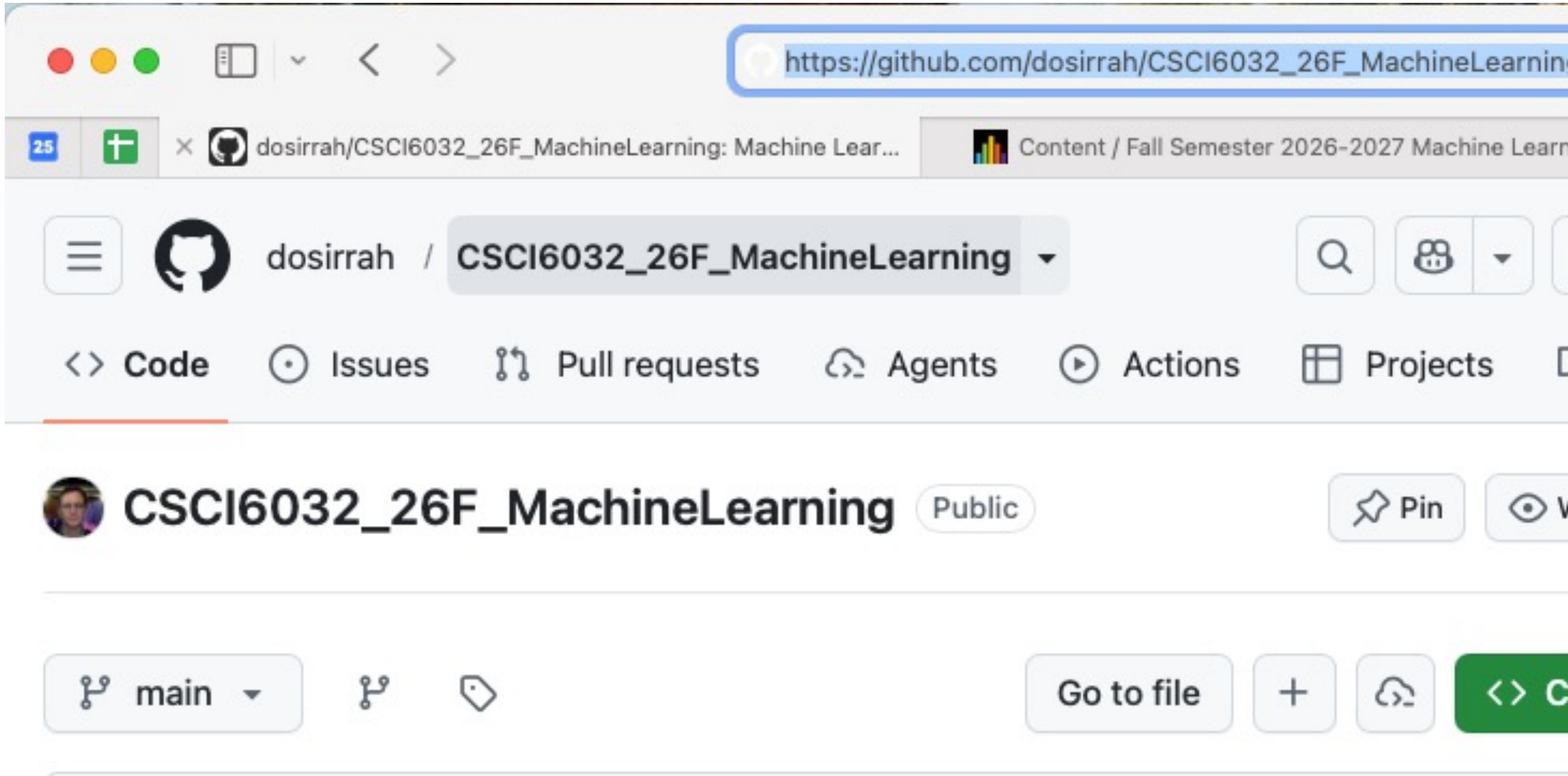
Notebooks and source code examples will also be committed to GitHub.

https://github.com/dosirrah/CSCI6032_26F_MachineLearning

You will need to create a Github account independent of your olemiss accounts.

GitHub is free for our purposes.

I highly recommend committing any code you create to GitHub.



Google Colab

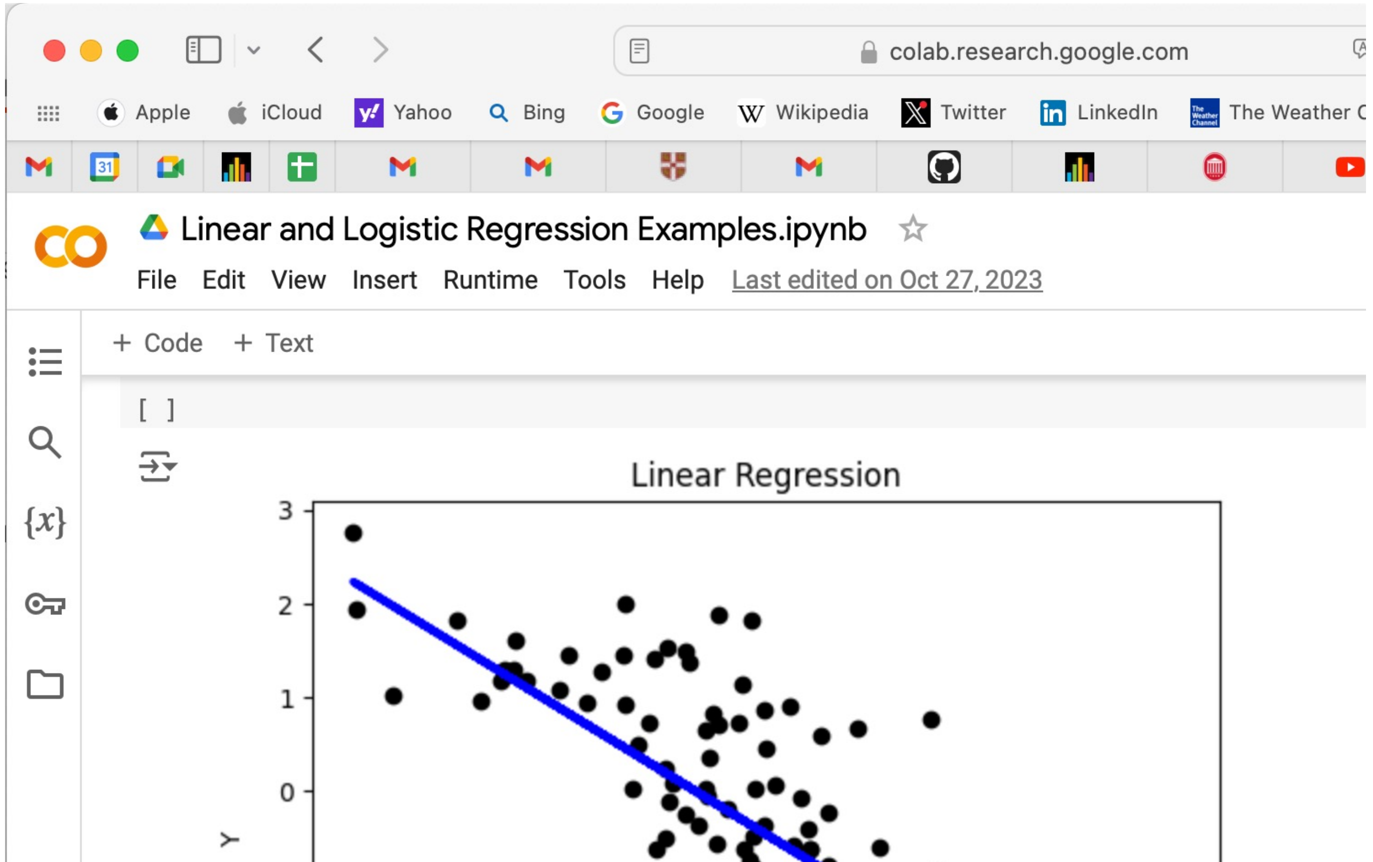
Notebooks and source code examples will also be committed to GitHub.

`https://github.com/dosirrah/CSCI6032_26F_MachineLearning`

Free.

Sufficient resources for the problems we will tackle.

All homeworks will either be scanned by hand or as a notebook.



Jupyter notebooks

If you like to run things locally. Jupyter notebooks.

In Mac OS, I recommend using Conda

`https://docs.anaconda.com/miniconda/`

Once conda is installed

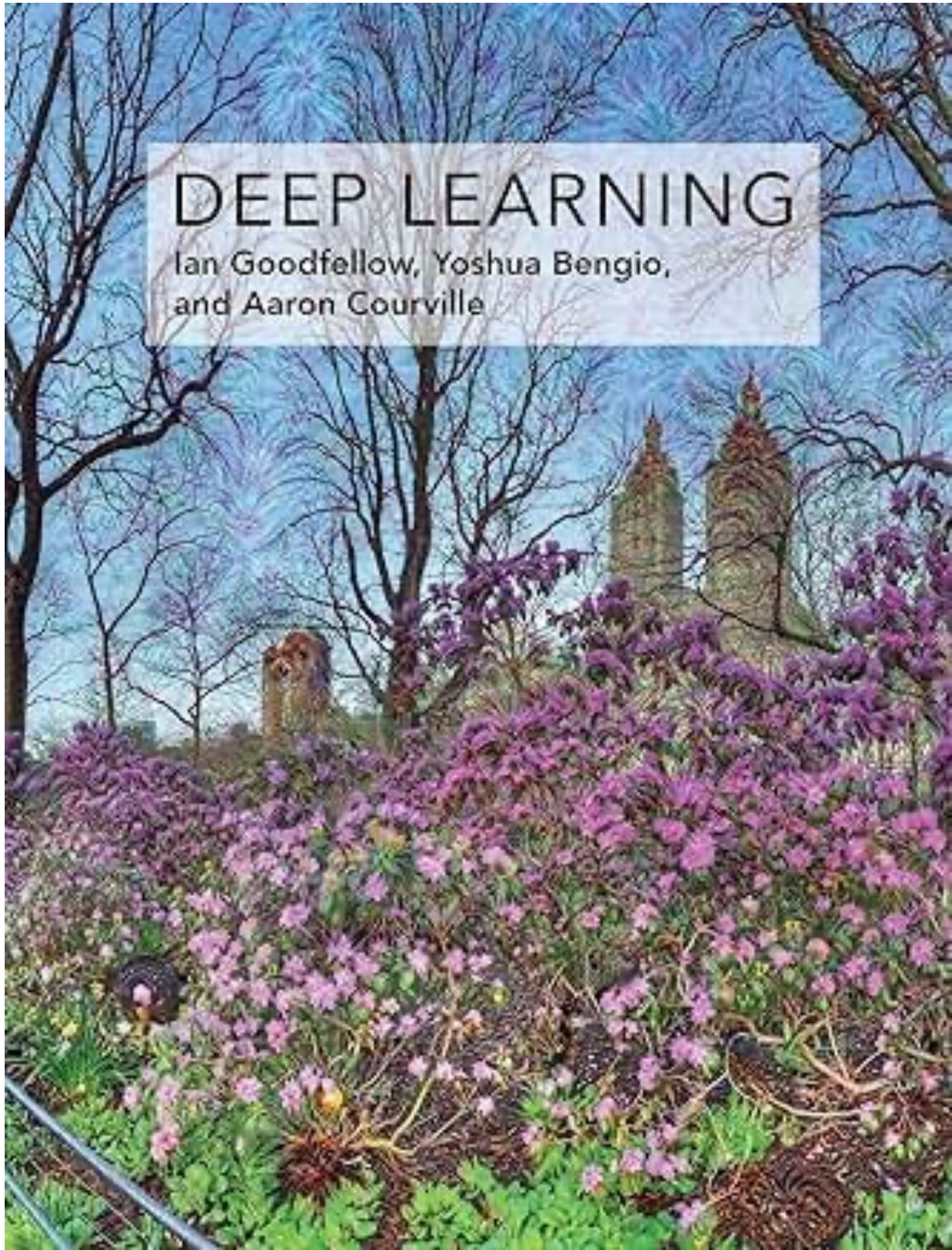
```
$ conda create –name myenv
$ conda activate myenv
$ conda install jupyter
```

Then to run jupyter notebook.

```
$ jupyter notebook
```

Class Textbook

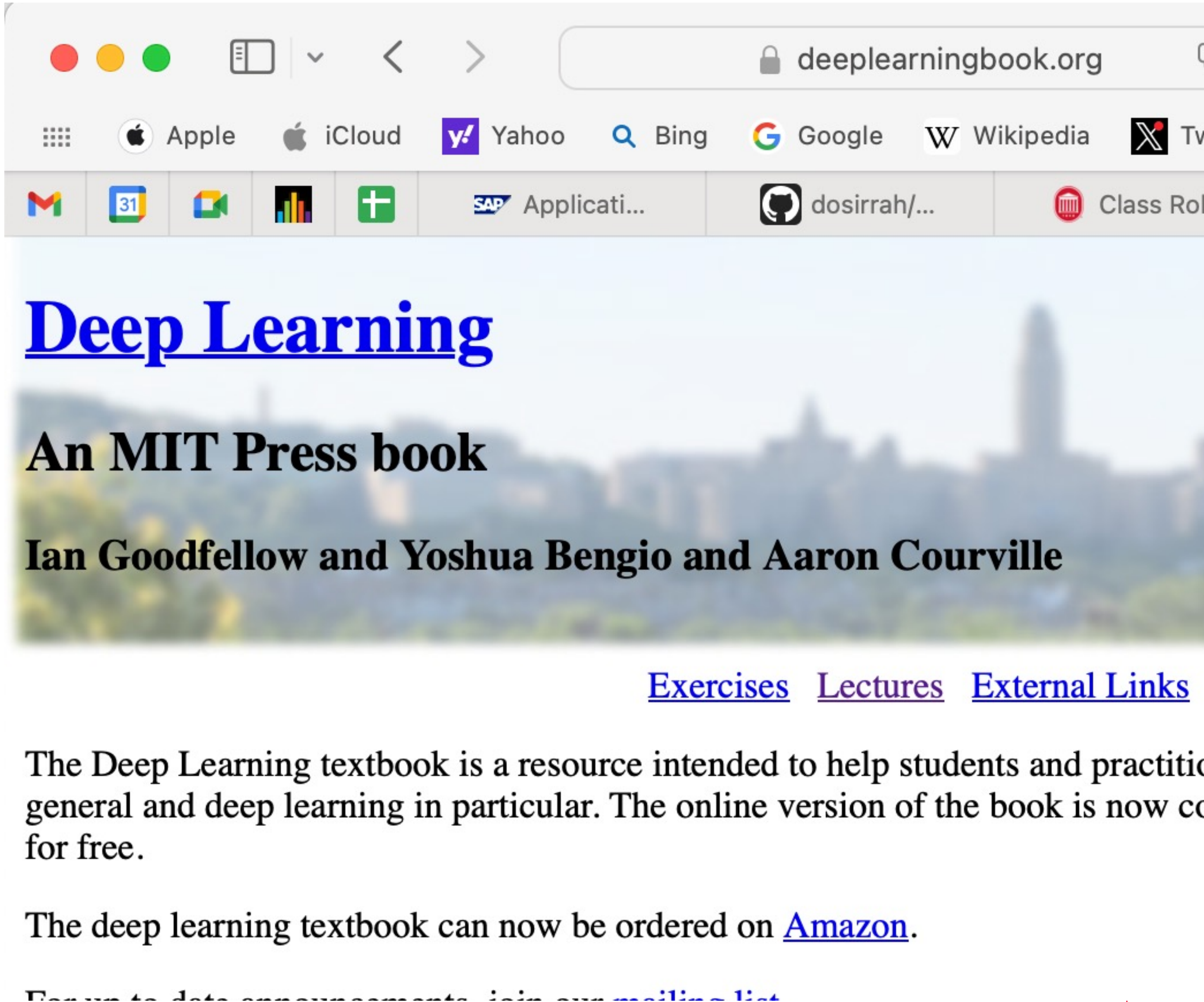
- *Deep Learning*, Ian Goodfellow, et al



Class Textbook

- ***Available online for free***

`deeplearningbook.org`

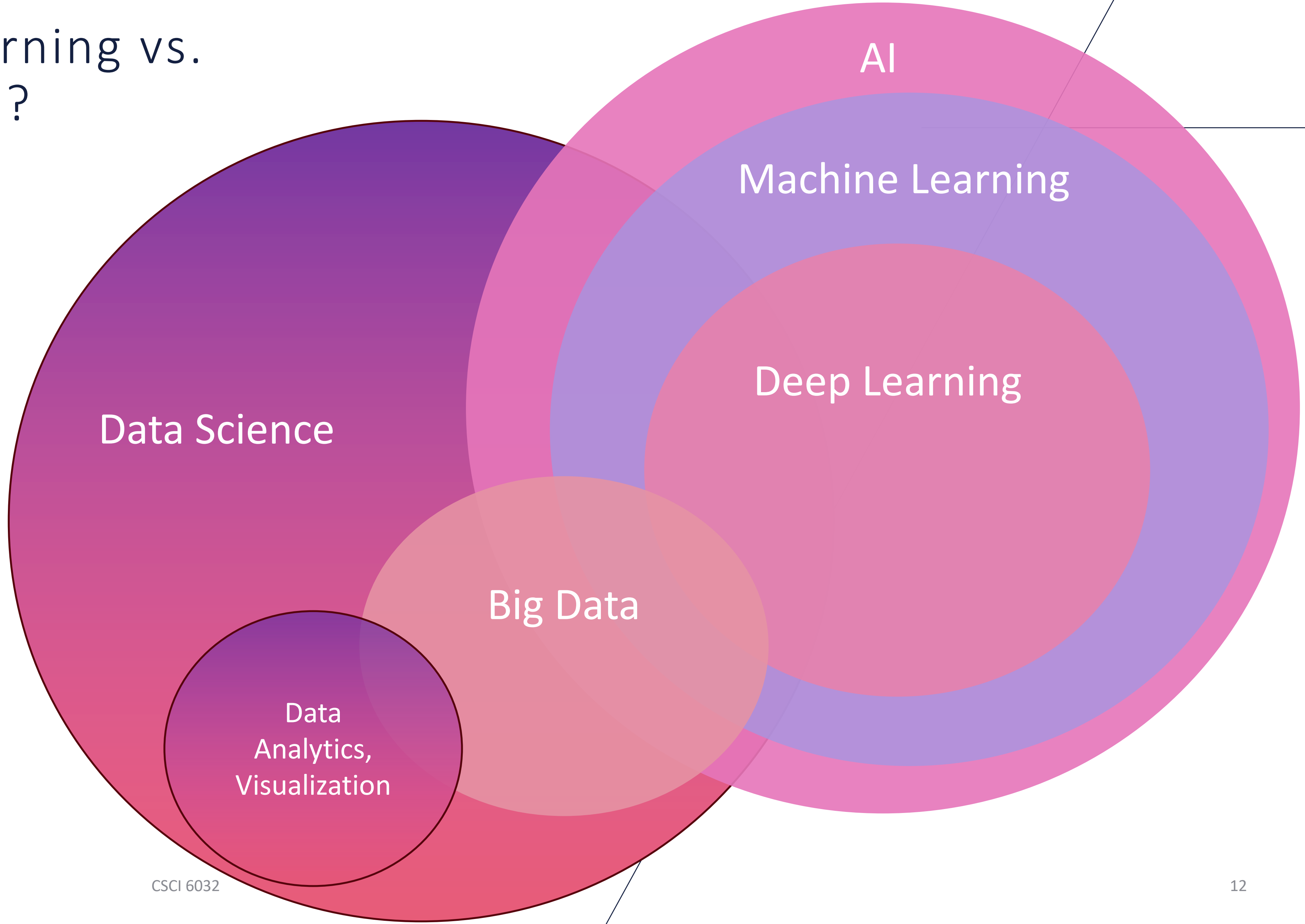


Machine learning vs. data science?

- **Wikipedia:**
  "Machine learning (ML) is a field of study in artificial intelligence concerned with the development and study of statistical algorithms that can learn from data and generalize to unseen data and thus perform tasks without explicit instructions."

  Goodfellow et al: "AI systems need the ability to acquire their own knowledge, by extracting patterns from raw data. This capability is known as machine learning."

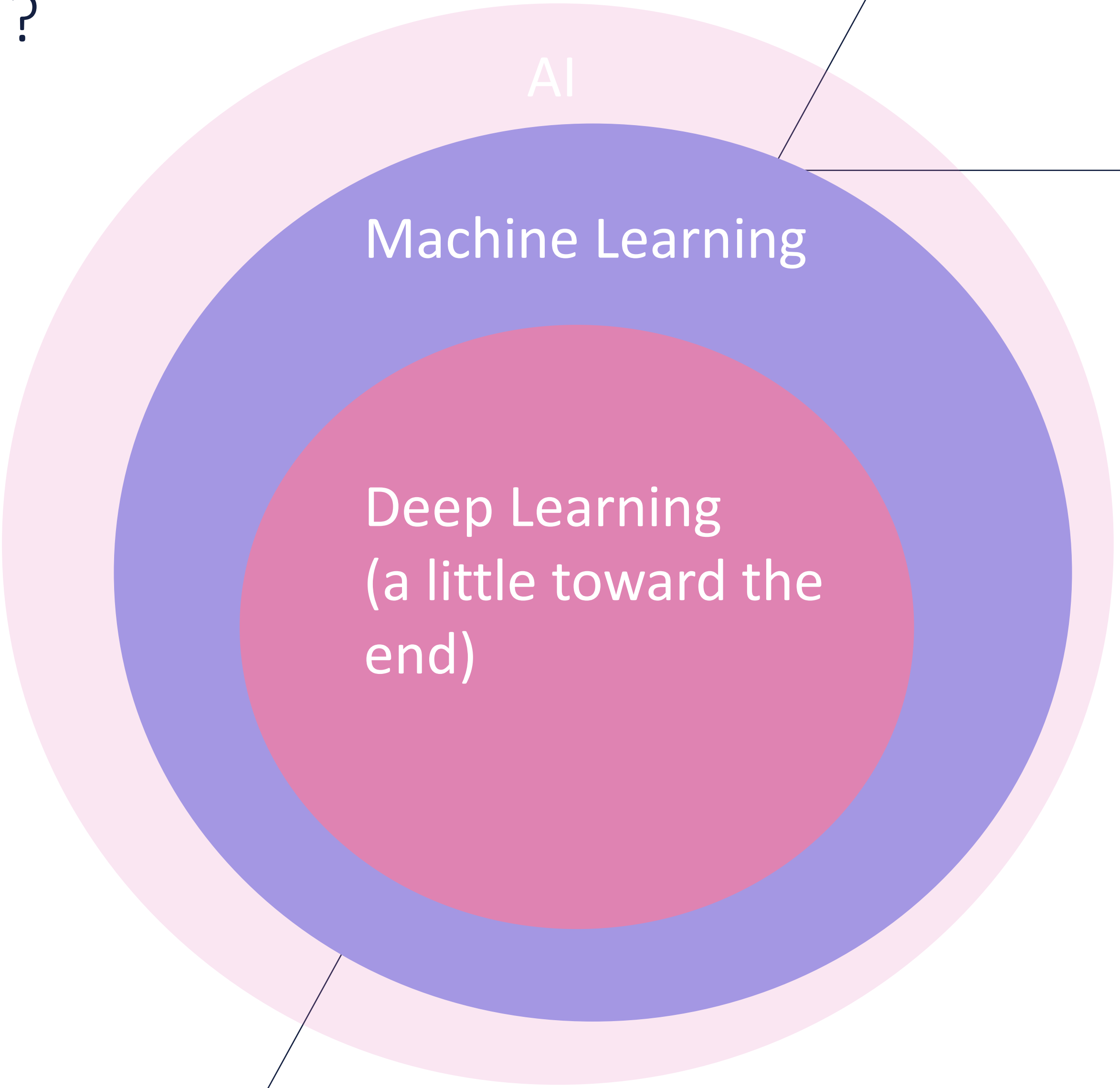
Machine learning vs. data science?



What is the focus of this course?

We cover a range of machine learning techniques.

Next semester there is a graduate course that focuses on deep learning.



Earliest machine learning example?

Claude Shannon in 1952 created a maze through which a magnetic mouse would navigate.

Inspired micromouse competitions.

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Inspired micromouse competitions.

Fortran wasn't invented until 1957.

Earliest learning in robotics?

This worked in two phases.

- Learning phase: magnetic mouse would use trial-and-error.
- Execution phase: put the mouse in maze again and it goes straight to the goal.

Examples of AI that do not use ML

Expert systems:

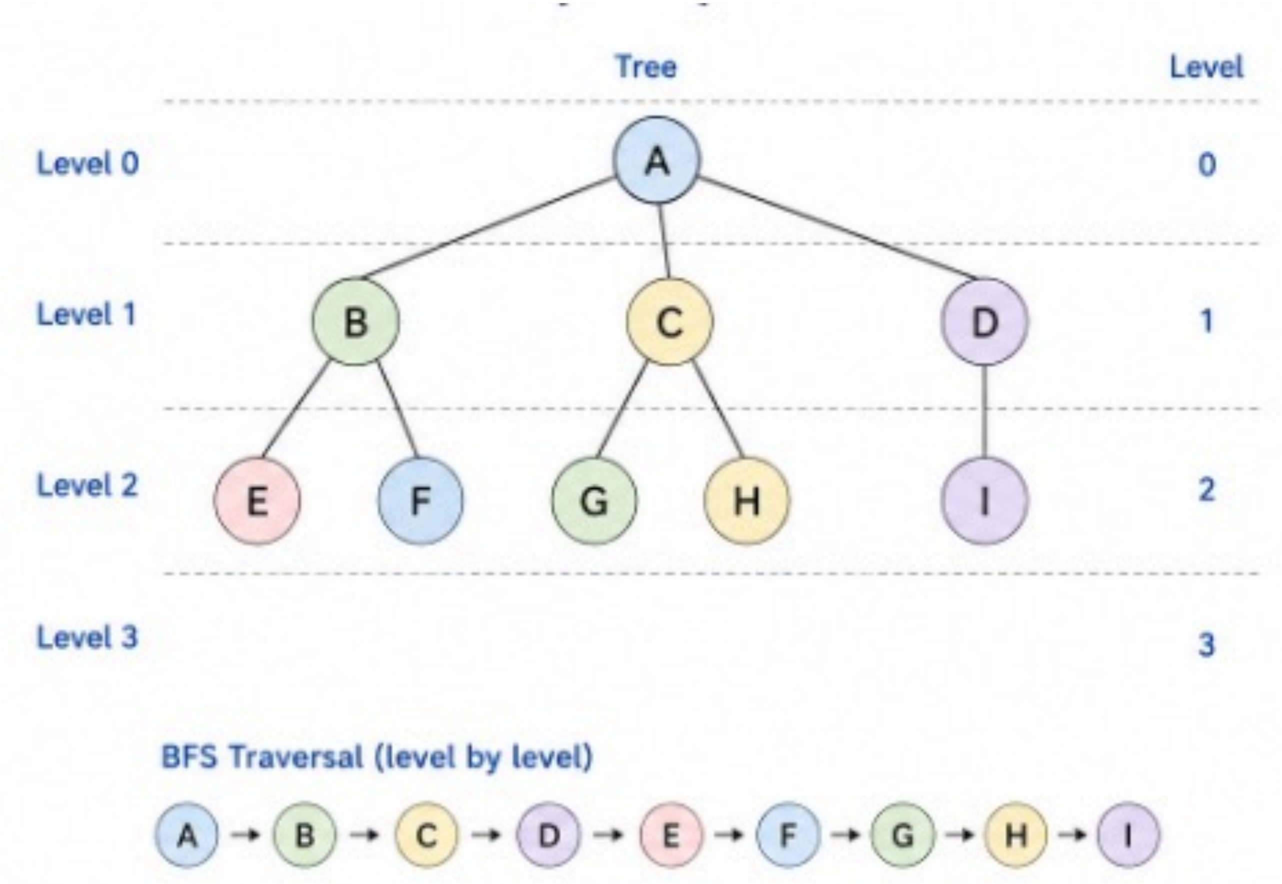
- Often use databases of questions with hard-coded answers.

```
IF fever
AND sore_throat
AND tonsillar_exudate
AND tender_cervical_nodes
AND NOT influenza
THEN suspect streptococcal_pharyngitis
```

Examples of AI that do not use ML

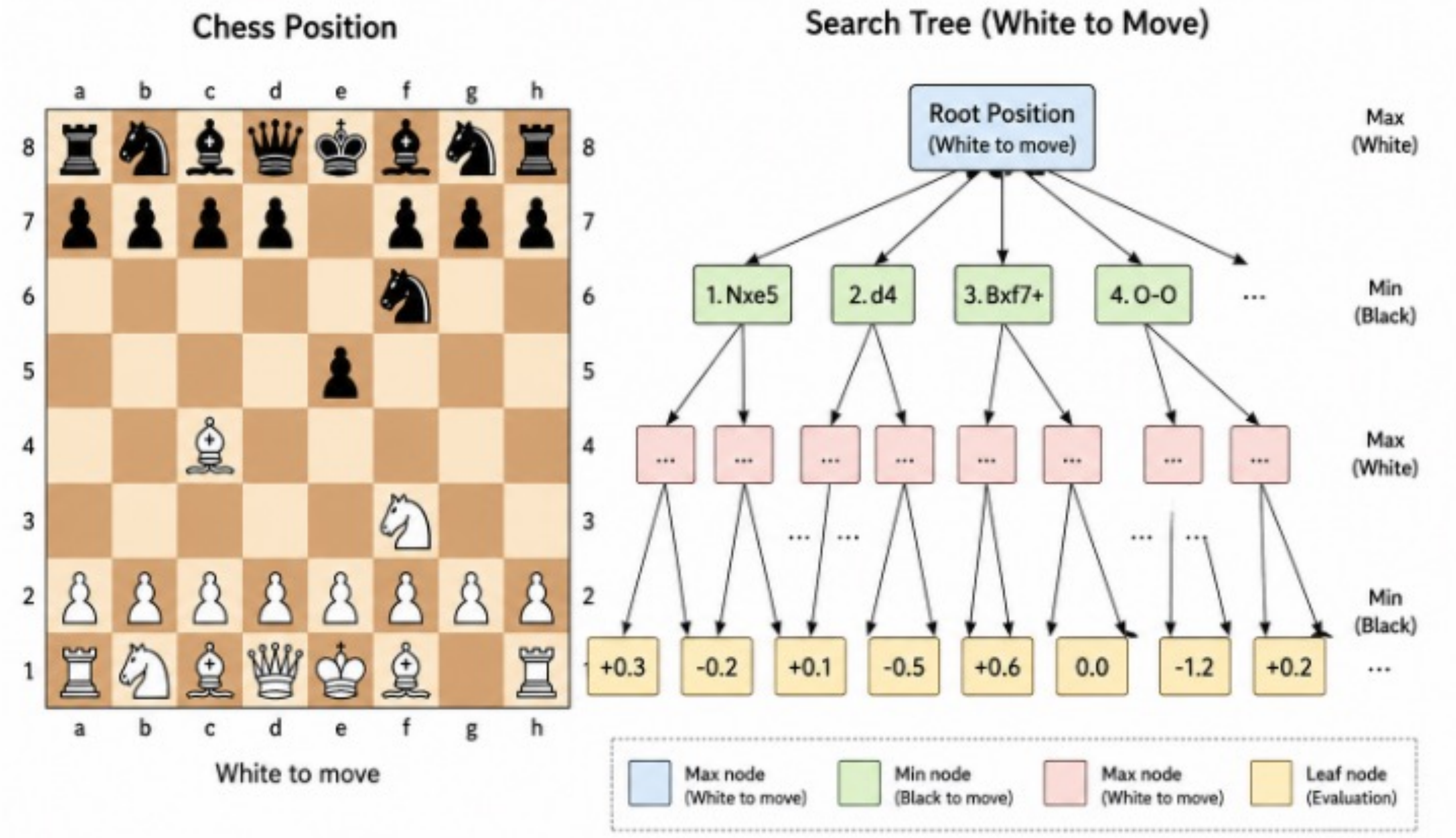
Search-based problem solving

- **Breadth first search**
- **Depth-first search**
- **A***
- **Minimax**
- **Alpha-beta prunig.**



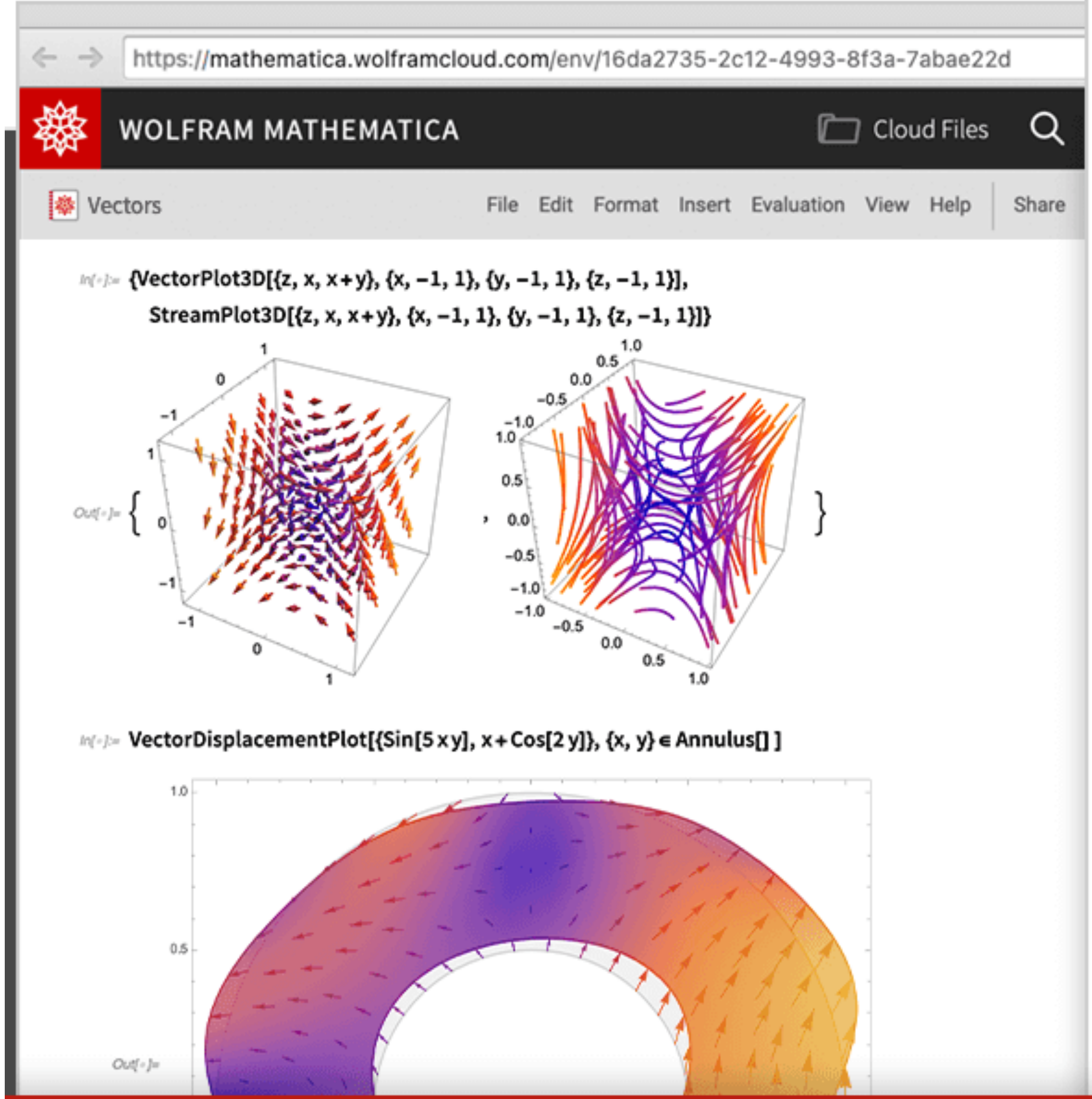
Examples of AI that do not use ML

Game playing (older)



Examples of AI that do not use ML

Symbolic computation



machine learning vs. data science?

Drew Conway's Venn Diagram

Data Science is machine learning with enough expertise to ask the right questions.